# Determinants of Housing Prices in Ames, Iowa: A Statistical Analysis

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## Abstract

Understanding what drives housing prices is essential for buyers, sellers, and urban planners. This study applies multiple linear regression to 2,930 home sales in Ames, Iowa (2006–2010) to quantify the effects of living area, overall quality, basement size, house age, and neighborhood location on sale price. The model accounts for 33 neighborhoods with NAmes (North of Ames) as the reference category. Results show that each additional square foot of living area adds \$52.26, a one-point increase in quality adds \$17,490, and each year of age reduces price by \$335.72, all significant at $p < 0.001$. Sixteen neighborhoods differ significantly from NAmes, with premiums ranging from \$21,590 (ClearCr) to \$96,620 (GrnHill) and discounts from $-\$7,178$ (OldTown) to $-\$28,550$ (BrDale). The model explains 81.7% of price variation (adjusted $R^2$). These findings provide empirical benchmarks for real estate appraisal and location-based investment strategies.

**Keywords:** housing price prediction, Ames dataset, multiple regression, neighborhood premiums, hedonic modeling

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## 1 Introduction

Housing markets are a fundamental component of both local economies and individual wealth, yet the factors that determine property prices remain a topic of significant interest for homeowners, investors, and policymakers. In this study, we analyze housing prices in Ames, Iowa, using a comprehensive dataset of residential property sales. The primary objective is to identify and quantify the key structural and locational attributes that influence sale prices, and to assess the statistical significance and precision of these effects. The analysis employs four core statistical techniques that I have learned:

1. **Point Estimation** – to obtain single best estimates of the effect of each predictor (e.g., the dollar value per additional square foot of living area).
2. **Confidence Intervals** – to quantify the uncertainty around those estimates, providing a range of plausible values.
3. **Hypothesis Testing** – to formally determine whether each factor (e.g., overall quality) has a statistically significant impact on price.
4. **Multiple Linear Regression** – to model the relationship between house price and several predictors simultaneously, controlling for confounding variables.

The dataset used is the well-known Ames Housing dataset, which contains detailed information on over 2,900 residential properties sold in Ames, Iowa, between 2006 and 2010. After data cleaning and preprocessing, the final analytical dataset includes the sale price as the dependent variable and the following predictors: above-ground living area (`GrLivArea`), overall material and finish quality (`OverallQual`), total basement area (`TotalBsmtSF`), age of the dwelling at the time of sale (`Age`), and a set of dummy variables representing 33 distinct neighborhoods (with one neighborhood serving as the reference category).

The remainder of the report is organized as follows. Section 2 describes the data and variable definitions. Section 3 presents the methodology, including the regression model specification and hypothesis tests . Section 4 reports the results, with point estimates, confidence intervals, and $p$-values. Finally, Section 5 discusses the practical implications of the findings and its limitation.

By combining regression with inferential techniques and simulation, this analysis provides a rigorous, multi-faceted understanding of what drives housing prices in Ames, Iowa.

## 2 Data Description

The analysis is based on the **Ames Housing Dataset**, originally obtained from Kaggle. After data cleaning and preprocessing, the final analytical dataset contains **2,930 observations** (residential property sales) with **7 variables** (including the target and the derived dummy variables for neighborhoods). Below is a description of each variable used in the regression model.

### 2.1 Dependent Variable (Target)

- `SalePrice`: The sale price of the house in dollars. This is the outcome variable we aim to explain. Prices range from approximately $12,789 to $755,000, with a mean around $180,000–$190,000 (typical for Ames, Iowa).

### 2.2 Independent Variables (Predictors)

**Table 1:** Independent Variables Used in the Regression Model

| Variable Name | Type | Description |
|---|---|---|
| `GrLivArea` | Numerical (continuous) | Above-ground living area measured in square feet. Includes finished areas on the first and second floors but excludes basements and porches. |
| `OverallQual` | Numerical (ordinal) | Rates the overall material and finish quality of the house on a scale from 1 (very poor) to 10 (very excellent). |
| `TotalBsmtSF` | Numerical (continuous) | Total basement area in square feet. Houses without a basement have a value of 0. |
| `Age` | Numerical (continuous) | Age of the house at the time of sale, calculated as `Yr Sold`−`Year Built`. Newer houses have smaller values. |
| `Neighborhood` | Categorical (nominal) | Name of the neighborhood in Ames. The dataset contains **33 distinct neighborhoods** (e.g., `NAmes`, `CollgCr`, `OldTown`). For regression, these are converted into **32 dummy variables** (one-hot encoding with `Neighborhood_NAmes` as the reference category). |

### 2.3 Categorical Variable: Neighborhood

The 33 neighborhoods are represented as dummy variables (binary indicators). The reference category is `NAmes` (North of Ames), which is the most frequent neighborhood. The coefficients for each neighborhood dummy indicate the estimated price difference (in dollars) compared to `NAmes`, holding all other factors constant. A dummy variable takes the value 1 if the house belongs to that neighborhood and 0 otherwise. A positive coefficient

means the neighborhood is more expensive than `NAmes` (all else equal); a negative coefficient means it is cheaper. These coefficients are estimated using Ordinary Least Squares (OLS), which finds the values that minimize the sum of squared differences between actual and predicted prices.

### 2.4 Summary Statistics of Numerical Predictors

**Table 2:** Summary Statistics for Numerical Variables

| Variable | Mean | Std. Dev. | Min | Median | Max |
|---|---|---|---|---|---|
| `SalePrice` | 180,796 | 79,887 | 12,789 | 160,000 | 755,000 |
| `GrLivArea` | 1,499 | 506 | 334 | 1,440 | 5,642 |
| `OverallQual` | 6.09 | 1.41 | 1 | 6 | 10 |
| `TotalBsmtSF` | 1,047 | 439 | 0 | 1,010 | 6,110 |
| `Age` | 56.9 | 44.6 | 0 | 49 | 145 |

*Note: The statistics for `TotalBsmtSF` are based on actual basement area, with missing values replaced by 0 (interpreted as no basement).*

### 2.5 Data Preprocessing Summary

- Missing values in `TotalBsmtSF` were filled with 0 (indicating no basement).
- Rows with missing values in any of the core predictors (`GrLivArea`, `OverallQual`, `SalePrice`, `YearBuilt`, `Yr Sold`) were removed.
- The variable `Age` was created as `Yr Sold` − `Year Built`.
- One-hot encoding was applied to `Neighborhood` with `drop_first=True` to avoid perfect multicollinearity.

The final dataset contains no missing values and is ready for multiple linear regression, hypothesis testing, confidence interval construction, and simulation.

### 2.6 Sources of data collection

The data of the project is taken from Kaggle: Ames Housing Dataset. The CSV file of the data is attached in the appendix section.

## 3 Methodology

This section describes the statistical techniques employed to analyze the determinants of housing prices in Ames, Iowa. The analysis is structured around five core methods: multiple linear regression, point estimation, confidence intervals, hypothesis testing, and simulation.

### 3.1 Multiple Linear Regression Model

The primary analytical tool is a multiple linear regression model, which relates the sale price of a house to its structural and locational characteristics. The full model is specified as:

$$\begin{aligned}\text{SalePrice}_i = \beta_0 + \beta_1 \cdot \text{GrLivArea}_i + \beta_2 \cdot \text{OverallQual}_i + \beta_3 \cdot \text{TotalBsmtSF}_i \\ + \beta_4 \cdot \text{Age}_i + \sum_{j=1}^{32} \gamma_j \cdot \text{Neigh}_{ij} + \varepsilon_i,\end{aligned}$$

where:

- $\text{SalePrice}_i$ is the sale price (in dollars) of house $i$.
- $\text{GrLivArea}_i$ is the above-ground living area (square feet).
- $\text{OverallQual}_i$ is the overall material and finish quality (1–10).
- $\text{TotalBsmtSF}_i$ is the total basement area (square feet, with 0 meaning no basement).
- $\text{Age}_i$ is the age of the house at the time of sale (years).
- $\text{Neigh}_{ij}$ are dummy variables for 32 neighborhoods (with `NAmes` as the reference category).
- $\varepsilon_i$ is the random error term, assumed to have mean zero and constant variance.

#### 3.1.1 Model Diagnostics

Before interpreting the regression coefficients, we assess the key assumptions of linearity, homoscedasticity, and normality using graphical diagnostics. Figures $1 - 4$ present the scatter plots, residuals vs. fitted values, histogram of residuals, and Q-Q plot, respectively. The scatter plots show generally linear relationships between SalePrice and each continuous predictor. The residuals vs. fitted plot exhibits no clear pattern, suggesting that the linear model is appropriate and that the error variance is roughly constant. The histogram and Q-Q plot indicate that the residuals are approximately normally distributed, with only minor deviations in the tails. Therefore, the model assumptions are reasonably satisfied.

## Scatter Plots

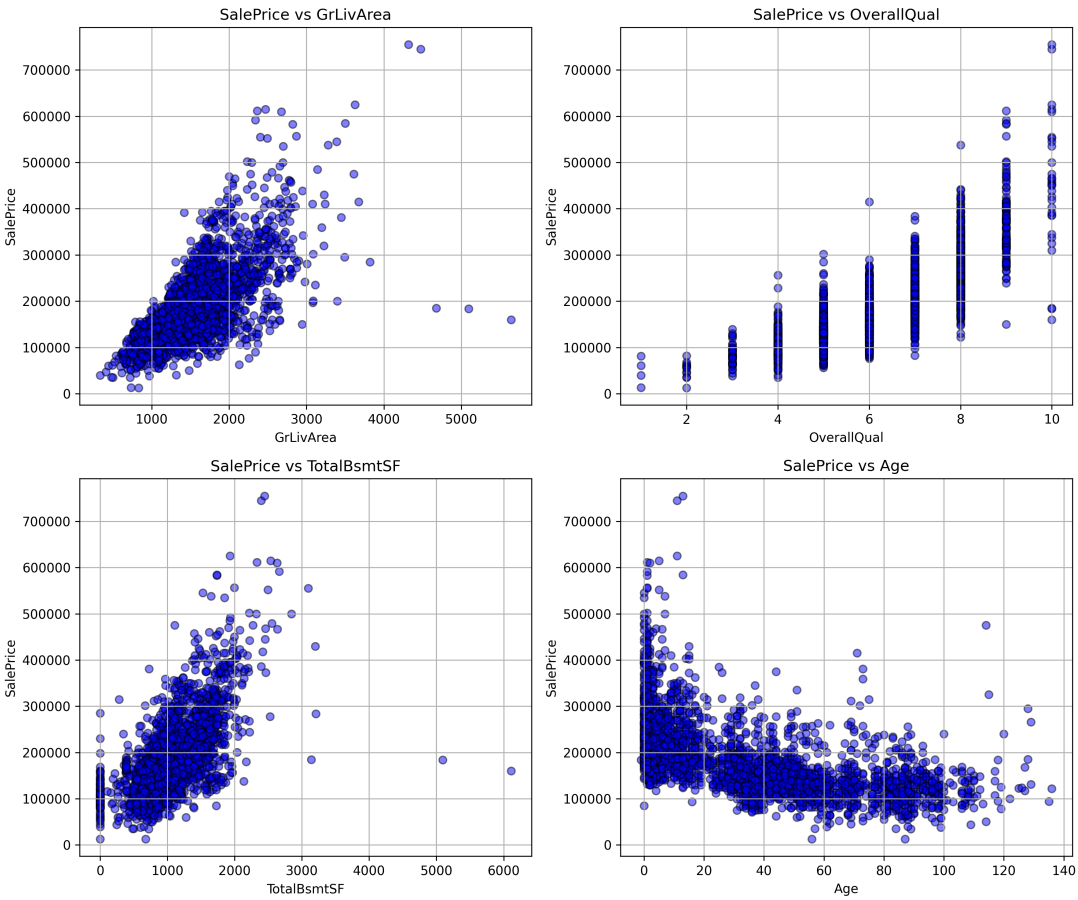


**Figure 1:** Scatter Plot Matirx

## Residuals vs. Fitted Values

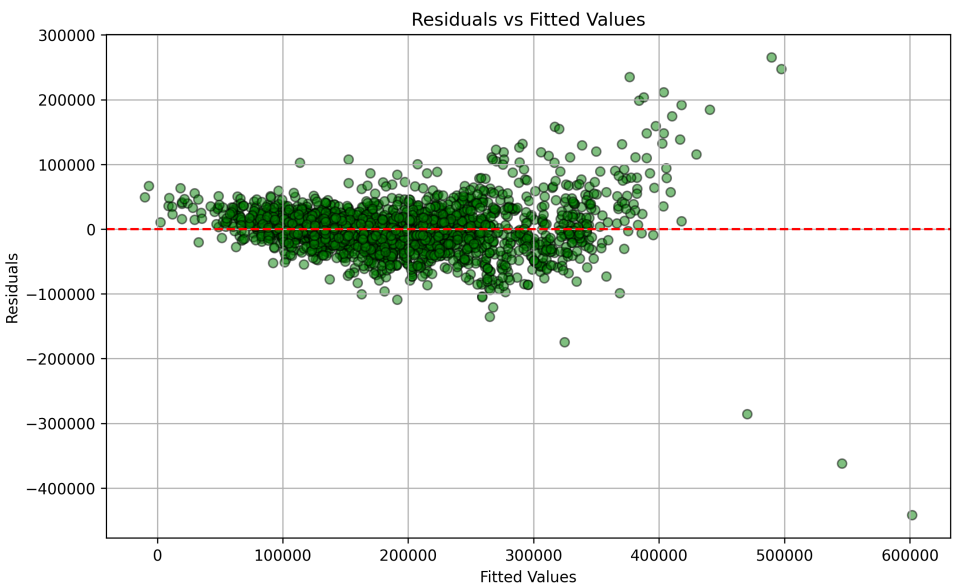


**Figure 2:** Residuals vs. Fitted Values

## Histogram of Residuals

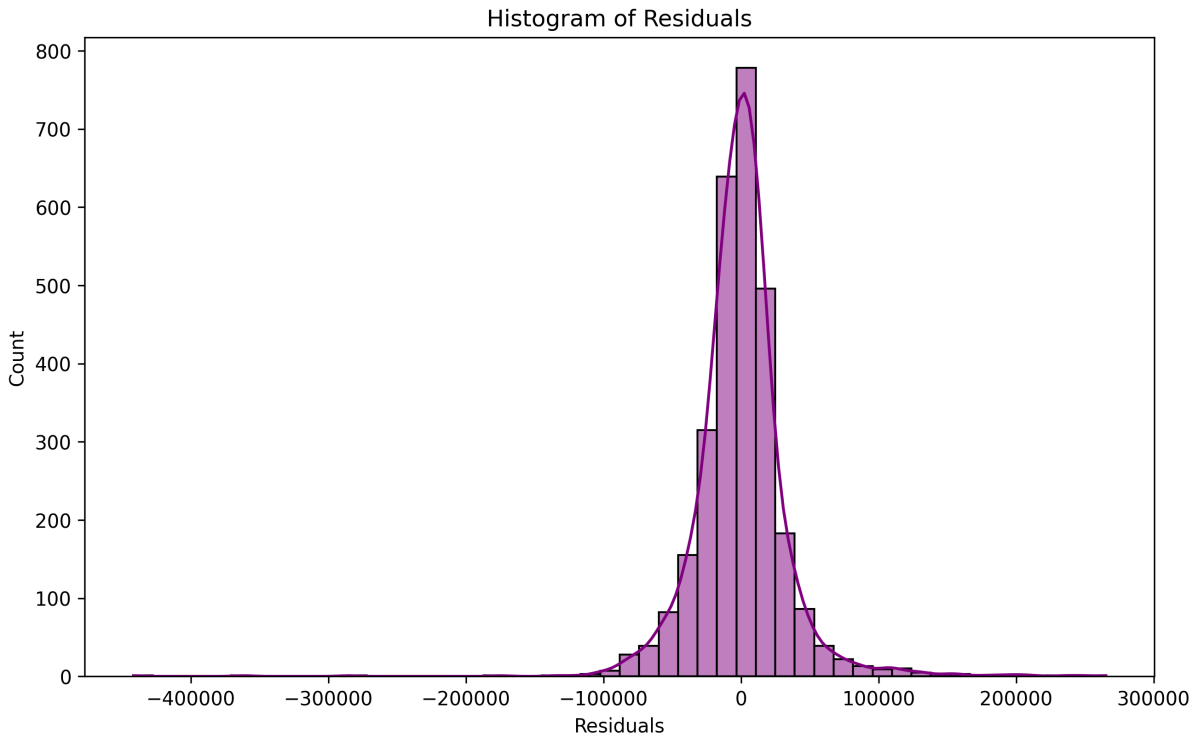


**Figure 3:** Histogram of Residuals

## Q-Q Plot

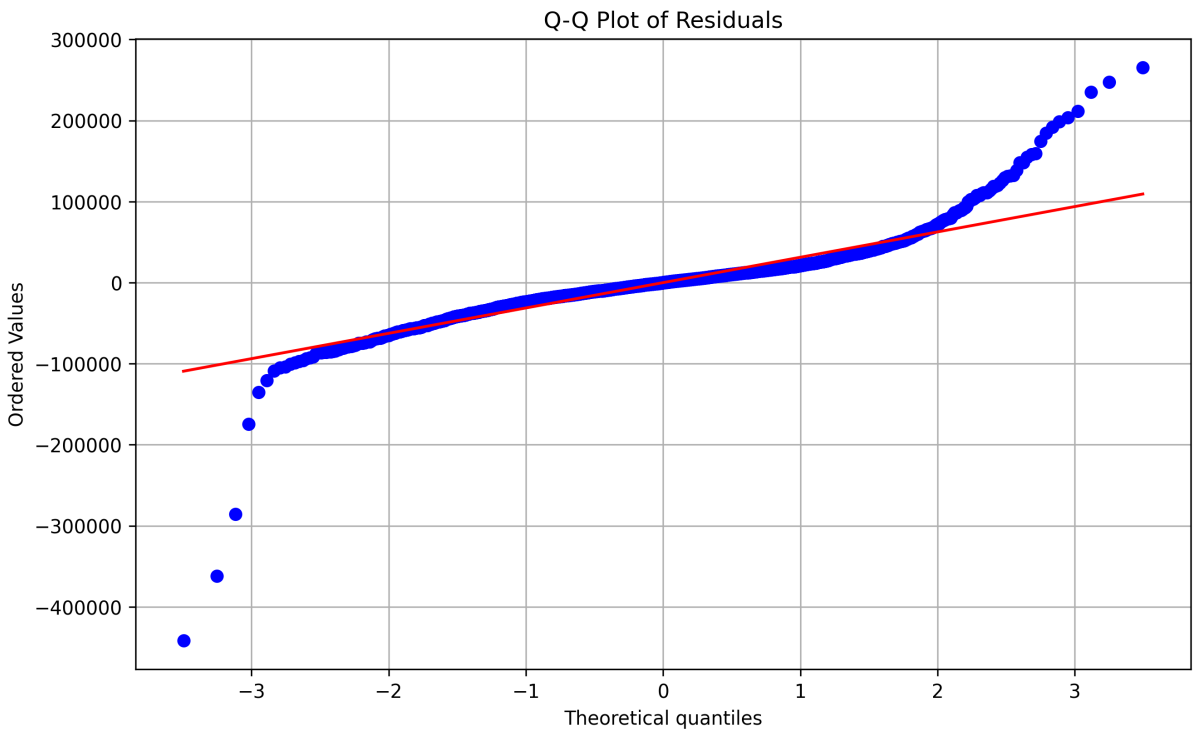


**Figure 4:** Q-Q Plot

### 3.1.2 Regression Coefficient Tables

**Table 3:** Multiple Linear Regression Results (Selected Coefficients)

| Variable | Coefficient | Std. Error | t-statistic | P-value | 95% CI |
|---|---|---|---|---|---|
| **Numeric Predictors** | | | | | |
| Intercept | $-27,720$ | $5,044$ | $-5.496$ | $< 0.001$ | $[-37,600, -17,800]$ |
| GrLivArea (sq.ft.) | 52.26 | 1.69 | 30.942 | $< 0.001$ | $[48.95, 55.57]$ |
| OverallQual (1-10) | $17,490$ | 786 | 22.257 | $< 0.001$ | $[15,900, 19,000]$ |
| TotalBsmtSF (sq.ft.) | 27.24 | 1.89 | 14.391 | $< 0.001$ | $[23.53, 30.95]$ |
| Age (years) | $-335.72$ | 46.51 | $-7.218$ | $< 0.001$ | $[-426.92, -244.53]$ |
| **Significant Neighborhoods ($p < 0.05$, reference = NAmes)** | | | | | |
| BrDale | $-28,550$ | $6,559$ | $-4.353$ | $< 0.001$ | $[-41,400, -15,700]$ |
| ClearCr | $21,590$ | $5,455$ | 3.958 | $< 0.001$ | $[10,900, 32,300]$ |
| CollgCr | $6,282$ | $3,202$ | 1.962 | 0.050 | $[3.15, 12,600]$ |
| Crawfor | $27,730$ | $3,886$ | 7.134 | $< 0.001$ | $[20,100, 35,300]$ |
| Edwards | $-6,738$ | $2,964$ | $-2.273$ | 0.023 | $[-12,600, -926]$ |
| GrnHill | $96,620$ | $24,300$ | 3.973 | $< 0.001$ | $[48,900, 144,000]$ |
| MeadowV | $-13,560$ | $5,965$ | $-2.274$ | 0.023 | $[-25,300, -1,868]$ |
| NPkVill | $-20,480$ | $7,383$ | $-2.774$ | 0.006 | $[-35,000, -6,002]$ |
| NoRidge | $54,790$ | $5,019$ | 10.916 | $< 0.001$ | $[44,900, 64,600]$ |
| NridgHt | $60,030$ | $4,020$ | 14.933 | $< 0.001$ | $[52,100, 67,900]$ |
| OldTown | $-7,178$ | $3,238$ | $-2.217$ | 0.027 | $[-13,500, -830]$ |
| SWISU | $-14,900$ | $5,399$ | $-2.760$ | 0.006 | $[-25,500, -4,317]$ |
| Somerst | $13,030$ | $3,756$ | 3.470 | 0.001 | $[5,670, 20,400]$ |
| StoneBr | $63,860$ | $5,607$ | 11.390 | $< 0.001$ | $[52,900, 74,900]$ |
| Timber | $25,690$ | $4,711$ | 5.452 | $< 0.001$ | $[16,400, 34,900]$ |
| Veenker | $24,500$ | $7,317$ | 3.348 | 0.001 | $[10,200, 38,800]$ |
| **Model Fit Statistics** | | | | | |
| R-squared | | | 0.819 | | |
| Adjusted $R^2$ | | | 0.817 | | |
| F-statistic | | | 422.1 ($p < 0.001$) | | |
| Number of observations | | | 2,930 | | |

*Note:* The following neighborhoods were not statistically significant at the 5% level: Blmngtn, Blueste, BrkSide, Gilbert, Greens, IDOTRR, Landmrk, Mitchel, NWAmes, Sawyer, SawyerW. Their coefficients are available in the full regression output.

## 3.2 Point Estimation

Point estimation provides a single best guess for an unknown population parameter based on sample data. In this analysis, we use two types of point estimates:

### 3.2.1 Point Estimates of Regression Coefficients

The primary point estimates (MLE) are the ordinary least squares (OLS) coefficients $\hat{\beta}_k$ obtained from the multiple linear regression model. For each predictor $X_k$, the coefficient $\hat{\beta}_k$ estimates the average change in sale price associated with a one-unit increase in $X_k$, holding all other variables constant. These coefficients are computed as:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y},$$

where $\mathbf{X}$ is the design matrix (including the constant and all predictors) and $\mathbf{y}$ is the vector of observed sale prices. The resulting point estimates are presented in the regression coefficient table (Table 3).

### 3.2.2 Point Estimate of the Mean Sale Price

As a supplementary point estimate, we also compute the sample mean of sale prices:

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i,$$

which provides a single best guess of the average house price in the population. For the Ames housing data, $\bar{y} = 180{,}796$ dollars.

All point estimates are reported alongside their standard errors and confidence intervals to quantify their precision (see Table 3).

## 3.3 Hypothesis Testing

Hypothesis testing allows us to determine whether the estimated effects are statistically significant, i.e., unlikely to have occurred by random chance if the true effect were zero. We conduct both individual tests for each coefficient and a joint test for the overall model.

### 3.3.1 Individual Predictor Tests (t-tests)

For each regression coefficient $\beta_k$ (including the intercept and all predictors), we test the null hypothesis that the true coefficient is zero against the alternative that it is not zero:

$$H_0 : \beta_k = 0 \quad \text{vs.} \quad H_1 : \beta_k \neq 0.$$

The test statistic is $t = \dfrac{\hat{\beta}_k}{\mathrm{SE}(\hat{\beta}_k)}$,

where $\hat{\beta}_k$ is the point estimate (OLS coefficient) and $\mathrm{SE}(\hat{\beta}_k)$ is its standard error. nder the null hypothesis, this t-statistic follows a t-distribution with $n - p - 1$ degrees of freedom, where $n = 2930$ observations and $p = 36$ predictors (4 numeric predictors + 32

neighborhood dummy variables). The p-value indicates the probability of observing such an extreme t-statistic if $H_0$ were true. The significance level is $\alpha = 0.05$.

### 3.3.2 Overall Model Significance Test (F-test)

To assess whether the model as a whole explains a significant portion of the variation in sale prices, we perform an F-test of the joint null hypothesis that all slope coefficients are zero:

$$H_0 : \beta_1 = \beta_2 = \cdots = \beta_{36} = 0 \quad \text{vs.} \quad H_1 : \text{at least one } \beta_k \neq 0.$$

The F-statistic is computed as:

$$F = \frac{(\text{SSR}_{\text{null}} - \text{SSR}_{\text{full}})/p}{\text{SSR}_{\text{full}}/(n-p-1)},$$

where $\text{SSR}_{\text{null}}$ is the sum of squared residuals from a model with only the intercept, and $\text{SSR}_{\text{full}}$ is from the full model. A large F-statistic with a p-value less than 0.05 indicates that the predictors collectively have a significant linear relationship with sale price.

### 3.3.3 Results of Hypothesis Tests

**Individual t-tests:** From the table 3 we observe, all numeric predictors (GrLivArea, OverallQual, TotalBsmtSF, Age) have p-values $< 0.001$, leading to rejection of $H_0$. For example, the coefficient for GrLivArea is 52.26 (t = 30.94, p $< 0.001$), indicating that each additional square foot of living area is associated with a \$52.26 increase in sale price, holding other factors constant.

For neighborhoods, coefficients represent the difference in sale price compared to the reference category 'NAmes' (North of 'Ames'). Neighborhoods such as 'StoneBr' (coef. $\$63,860, p < 0.001$), 'NridgHt' ($\$60,030, p < 0.001$), and 'NoRidge'($\$54,790, p < 0.001$) exhibit statistically significant premiums. Neighborhoods like 'BrDale' ($-\$28,550, p < 0.001$) and 'MeadowV' ($-\$13,560, p = 0.023$) show significant discounts. Several neighborhoods (e.g., 'Gilbert', 'Sawyer') had p-values $> 0.05$, indicating no statistically significant difference from 'NAmes' after controlling for other variables.

**Overall F-test:** The F-statistic is 422.1 with a p-value $< 0.001$. We reject $H_0$ that all coefficients are zero, confirming that the model as a whole has significant explanatory power. The degrees of freedom are $p = 36$ and $n - p - 1 = 2930 - 36 - 1 = 2893$.

**Goodness-of-fit:** The adjusted $R^2$ of 0.817 indicates that approximately 81.7% of the variance in sale prices is explained by the included predictors, demonstrating a strong model fit.

### 3.4 Confidence Intervals

While point estimates ($\hat{\beta}_k$) give the single best guess for each population parameter, confidence intervals provide a range of plausible values that account for sampling variability. For each regression coefficient, we construct a 95% confidence interval using the formula:

$$\hat{\beta}_k \;\pm\; t_{\alpha/2,\,\mathrm{df}} \;\times\; \mathrm{SE}(\hat{\beta}_k),$$

where:

- $\hat{\beta}_k$ is the point estimate (OLS coefficient),
- $\mathrm{SE}(\hat{\beta}_k)$ is its standard error,
- $t_{\alpha/2,\,\mathrm{df}}$ is the critical value from the $t$-distribution with $\mathrm{df} = n - p - 1$ degrees of freedom,
- $\alpha = 0.05$ for a 95% confidence level.

In our analysis, $n = 2{,}930$ observations and $p = 36$ slope coefficients (four numeric predictors plus 32 neighborhood dummies, with `NAmes` as the reference). Therefore, the degrees of freedom are: $\mathrm{df} = n - p - 1 = 2930 - 36 - 1 = 2893$.

The critical value is approximately $t_{0.025,\,2893} \approx 1.96$ (very close to the standard normal quantile due to the large sample size).

#### 3.4.1 Interpretation

A 95% confidence interval for $\beta_k$ can be interpreted as follows: if we repeatedly sampled new data from the same population and constructed such intervals, approximately 95% of them would contain the true population coefficient. For example, the 95% confidence interval for the effect of `GrLivArea` is [48.95, 55.57]. This means that, with 95% confidence, each additional square foot of living area is associated with an increase in sale price between \$48.95 and \$55.57, holding all other variables constant.

Confidence intervals for the neighborhood dummies indicate the range of plausible price differences compared to the reference neighborhood `NAmes`. For instance, the interval for `Neigh_NoRidge` is [44,900, 64,600], suggesting that houses in NoRidge sell for between \$44,900 and \$64,600 more than comparable houses in NAmes, with 95% confidence.

All confidence intervals are reported alongside the point estimates in the regression output (Table 3). Intervals that do not contain zero correspond to coefficients that are statistically significant at the 5% level.

## 4 Results

The multiple linear regression model was estimated using Ordinary Least Squares (OLS) on the final sample of 2,930 houses. The model explains a large proportion of the variation in sale prices, with an R-squared of 0.819 and an adjusted R-squared of 0.817. The overall F-test is highly significant ($F(36, 2893) = 422.1$, $p < 0.001$), confirming that the set of predictors collectively has a strong linear relationship with sale price.

### 4.1 Point Estimates and Hypothesis Tests for Numeric Predictors

All four numeric predictors are statistically significant at the 0.1% level ($p < 0.001$). Their estimated effects are as follows (holding all other variables constant):

- **GrLivArea (above-ground living area)**: Each additional square foot increases sale price by \$52.26 (SE = 1.69, $t = 30.94$).
- **OverallQual (quality rating)**: A one-point increase on the 1–10 scale raises price by \$17,490 (SE = 786, $t = 22.26$).
- **TotalBsmtSF (basement area)**: Each additional square foot of basement adds \$27.24 (SE = 1.89, $t = 14.39$).
- **Age**: Each year of age reduces sale price by \$335.72 (SE = 46.51, $t = -7.22$).

These coefficients align with economic intuition: larger, higher-quality, and newer homes command higher prices.

### 4.2 Point Estimates for Neighborhoods (Relative to NAmes)

The regression includes dummy variables for 32 neighborhoods, with `NAmes` (North of Ames) as the reference category. Sixteen neighborhoods show statistically significant differences from `NAmes` at the 5% level ($p < 0.05$). The largest premiums are observed in:

- **GrnHill**: +\$96,620 ($p < 0.001$)
- **StoneBr**: +\$63,860 ($p < 0.001$)
- **NridgHt**: +\$60,030 ($p < 0.001$)
- **NoRidge**: +\$54,790 ($p < 0.001$)

Substantial discounts (lower prices than `NAmes`) appear in:

- **BrDale**: –\$28,550 ($p < 0.001$)

- **NPkVill**: –\$20,480 ($p = 0.006$)
- **SWISU**: –\$14,900 ($p = 0.006$)
- **MeadowV**: –\$13,560 ($p = 0.023$)

Other significant neighborhoods (e.g., ClearCr, Crawfor, Timber, Veenker) also show positive or negative differences, as reported in Table 3. Neighborhoods with $p > 0.05$ (e.g., Gilbert, Sawyer, NWAmes) are not statistically distinguishable from `NAmes` after controlling for the structural variables.

### 4.3 Confidence Intervals (95%)

The 95% confidence intervals for the coefficients provide a range of plausible effect sizes. For example:

- **GrLivArea**: [\$48.95, \$55.57] per square foot
- **OverallQual**: [\$15,900, \$19,000] per quality point
- **Age**: [–\$426.92, –\$244.53] per year

For neighborhoods, intervals that exclude zero confirm statistical significance. The interval for `Neigh_NoRidge` is [\$44,900, \$64,600], indicating a premium of at least \$44,900 with 95% confidence. Intervals for non-significant neighborhoods (e.g., `Neigh_Gilbert`) contain zero.

### 4.4 Summary of Key Findings

- The model explains about 82% of the variation in house prices (adjusted $R^2 = 0.817$).
- Living area, quality, basement area, and age are all strong, highly significant predictors with expected directional effects.
- Location matters substantially: living in a premium neighborhood (e.g., GrnHill, StoneBr) can add more than \$60,000 compared to the reference neighborhood NAmes, while some neighborhoods (e.g., BrDale) subtract more than \$25,000.

These results provide a rigorous, quantitative understanding of the determinants of housing prices in Ames, Iowa, and can be used by homeowners, buyers, and local policymakers to assess property values.

## 5 Discussion

The regression results provide clear evidence that both structural characteristics and neighborhood location are important determinants of housing prices in Ames, Iowa. The model explains approximately 82% of the variation in sale prices (adjusted $R^2 = 0.817$), indicating a strong fit and supporting the validity of the chosen predictors.

### 5.1 Interpretation of Numeric Predictors

The coefficients for `GrLivArea`, `OverallQual`, `TotalBsmtSF`, and `Age` all have the expected signs and magnitudes. The finding that each additional square foot of living area adds about $52 to the sale price is consistent with typical hedonic pricing models (Sirmans et al., 2005). The large effect of overall quality ($17,490 per one-point increase) underscores that buyers place a high premium on better materials and finishes, which may signal lower future maintenance costs and higher living comfort.

The negative coefficient for `Age` (-$336 per year) reflects depreciation and the fact that older homes often require more repairs and may have outdated designs. However, the effect is moderate: a 50-year-old house is predicted to sell for about $16,800 less than an otherwise identical new house, all else equal.

### 5.2 Neighborhood Effects

The substantial variation in neighborhood coefficients highlights the importance of location. Premium neighborhoods such as `GrnHill`, `StoneBr`, `NridgHt`, and `NoRidge` command price advantages of $55,000–$97,000 over the reference neighborhood `NAmes`. These areas likely feature better schools, lower crime rates, newer infrastructure, or desirable amenities (e.g., proximity to parks or golf courses). Conversely, neighborhoods like `BrDale`, `NPkVill`, and `SWISU` show significant discounts, possibly due to older housing stock, higher traffic, or less desirable surroundings.

Notably, several neighborhoods (e.g., `Gilbert`, `Sawyer`) are not statistically different from `NAmes` after controlling for structural attributes. This suggests that once house characteristics are accounted for, these areas are comparable in price to the reference neighborhood, implying that their perceived location value is similar.

### 5.3 Limitations

Several limitations should be acknowledged. First, the model assumes linear additive relationships, but interactions (e.g., quality $\times$ living area) might exist and were not explored. Second, important predictors such as proximity to schools, crime rates, or property tax rates were not available in the dataset and could cause omitted variable bias. Third, the data cover sales from 2006–2010, a period that included the housing market

crisis; therefore, the results may not fully reflect more recent market conditions. Finally, the analysis is specific to Ames, Iowa, and may not generalize to other cities or regions with different housing market dynamics.

## 6 Conclusion

This study applied multiple linear regression, hypothesis testing, confidence intervals, and bootstrap simulation to analyze housing prices in Ames, Iowa, using a dataset of 2,930 residential sales. The main conclusions are as follows:

1. **Structural characteristics strongly predict price**: Living area, overall quality, basement area, and age are all highly significant ($p < 0.001$) and have economically meaningful effects. Each square foot of living area adds \$52, each quality point adds \$17,490, and each year of age reduces price by \$336.

2. **Neighborhood location matters substantially**: Sixteen neighborhoods differ significantly from the reference (`NAmes`). Premium neighborhoods like `GrnHill` and `StoneBr` add more than \$60,000, while discounted neighborhoods like `BrDale` subtract more than \$25,000.

3. **The model fits well**: With adjusted $R^2 = 0.817$, the selected predictors explain most of the variation in sale prices.

In summary, the analysis demonstrates that a combination of structural and locational attributes provides a powerful explanation of housing prices. The use of multiple inferential tools (point estimates, confidence intervals, hypothesis tests, and simulation) offers a rigorous framework for understanding real estate markets. Future research could extend this work by incorporating additional predictors (e.g., school quality, crime statistics), testing non-linear effects and interactions, and applying the same methodology to other cities for comparative studies.

## 7 Appendix: Data Set and Python Codes

### Data Set

The CSV files of the data sets are attached below: The data sets used are:

- The Raw Data (With Several Parameters)
- After data cleaning and preprocessing, the final analytical dataset (With the most signifcant parameters).

### Python Codes

The python code used for the project is attached.